KCE PATHWAYS · GRADE 3 · UNIT 5: DATA AND DESIGN · WEEK 25 · CHECKPOINT

Checkpoint 5 — Data and Design Review — Teacher Lesson Plan & Answer Key

Learning objectives: Students show mastery of the Unit 5 math skills (picture graphs and scaled bar graphs; measuring to the half and quarter inch and line plots; quadrilaterals and equal parts; two-step problems with a letter for the unknown, and rounding) and reading skills (main idea and key details; text features and search tools; point of view; biography, sequence, and cause and effect), including two-step problems, "which explanation is correct?" reasoning and "find the mistake" items.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · inch ruler marked in quarter inches · printed Checkpoint Pre & Post sheet

Lesson flow (as it appears on screen)

1. Title and name: the student types first and last name so results go to KingCretot Experience. Read the Pathway Points rules together: 2 tries per question, points only for a correct first try, and the answer and reason appear after a second miss.
2. Warm-Up: A Quick Unit 5 Mix (6 pts, 3 each): one math item and one reading item from across the unit.
3. Picture It: The Unit 5 Display Board. Rosalind and Cyrus build a display board for the Data and Design Fair with 8 panels, one for each Unit 5 lesson (Pike Place Market graphs, the Monterey Bay kelp forest, the Ozark woodshop, Mammoth Cave, the Santa Fe tile workshop, the Louisiana bayou houseboat, the Kansas City barbecue festival and the Roberto Clemente ballpark biography). Ask the student to name one skill from each panel.
4. Vocabulary Round-Up: 8 flip cards from across the unit (key, line plot, rhombus, equation, main idea, index, narrator, biography). Have the student say each meaning before flipping.
5. Pencil Break (Pre): the student opens or prints the Checkpoint Pre/Post sheet (grade3/prepost/KCE_Checkpoint_PrePost_G3W25.pdf) and completes Page 1 (Pre) before the review. Do not help yet; it shows what they remember. Item 2 needs a ruler.
6. Math Review: Picture Graphs and Bar Graphs (Week 21, 6 pts). Tip: read the key or the scale first; half a picture is half the key.
7. Math Review: Measuring and Line Plots (Week 22, 6 pts). Tip: check where the object starts; on a line plot, each X is one thing measured.
8. Math Review: Quadrilaterals and Equal Parts (Week 23, 6 pts). Tip: a square is both a rectangle and a rhombus; equal parts need equal area.
9. Math Review: Two-Step Problems and Rounding (Week 24, 6 pts). Tip: find the hidden question first; do × or ÷ before + or −.
10. Reading Review: Main Idea and Key Details (Week 21, 6 pts): a short nonfiction text in each set. Ask: which sentence do ALL the others explain?
11. Reading Review: Text Features and Search Tools (Week 22, 6 pts): a field guide page in each set. Ask: do you need a page number or a meaning?

12. Reading Review: Point of View (Week 23, 6 pts): a short story in each set. Remind: skip the words inside quotation marks when you look for I or she.
13. Reading Review: Biography, Sequence, and Cause and Effect (Week 24, 6 pts): the same short Mae Jemison biography in each set. Tip: dates are the strongest order clues; so comes before an effect.
14. Match-Up Round (10 pts, 2 per first-try match): all three sets match Unit 5 math and reading vocabulary to meanings.
15. Fix the Sentence (8 pts, 4 each): typed answers are checked exactly (capital letters, names, city-and-state comma, end mark, list commas, and the one verb fix in Set 2 and Set 3); curly quotes and extra spaces are forgiven. Both serial-comma and no-serial-comma list forms are accepted.
16. Type-It Corral (8 pts, 4 each): one typed math answer and one typed reading word. Capital letters do not matter; spelling does.
17. Speed Arena and Keyboard Corral (XP only, not scored).
18. Boss Battle, Part 1: The Data and Design Fair (10 pts, 5 each): a flyer per set (the fair, the Tile Art Table, or the Reading Corner), then a two-step math item and a reasoning item about the flyer.
19. Boss Battle, Part 2: The Last Panel (10 pts, 5 each): a graph per set (a visitor bar graph, a tile frame line plot, or a book picture graph), then a find-the-mistake or "is it reasonable?" item.
20. Pencil Break (Post): the student turns to Page 2 (Post) of the Checkpoint sheet (grade3/prepost/KCE_Checkpoint_PrePost_G3W25.pdf) and completes it, plus the ★ Challenge if time allows, before seeing the score.
21. Results (Checkpoint 5 Complete!): the score out of 100 is sent to KingCretot Experience. 85 or higher shows "Unit 5 Champion!" with the congratulations message "Unit 5 complete! You mastered Data and Design!" and a big confetti burst; below 85 shows "Great Effort Today!" with encouragement ("Every great display board is built one panel at a time."), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: A QUICK UNIT 5 MIX (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
`w1_s0`	Set 1: A picture graph key says each star stands for 5 tickets. One row has 6 stars. How many tickets does that row show?	**30** (also accepted: 30 tickets)	Read the key first. Each star is 5 tickets, so skip-count by 5 six times: 5, 10, 15, 20, 25, 30. That is 6 × 5 = 30 tickets.
`w2_s0`	Set 1: Which words name the most important point of a paragraph? [the main idea / a key detail / a caption / the index]	**the main idea**	The main idea is the most important point. Key details are the facts that prove it.
`w1_s1`	Set 2: How many quarter inches make 2 whole inches?	**8** (also accepted: 8 quarter inches, 8 quarters, eight)	Each inch has 4 quarter inches. 2 inches have 4 + 4 = 8 quarter inches.
`w2_s1`	Set 2: A story is told in first person. Which clue words does the narrator use? [I, me, and my / he, she, and they / only names / because and so]	**I, me, and my**	A first-person narrator is a character in the story, so the narrator says I, me, and my.

ID	Question	Answer	Why (explain to student)
w1_s2	Set 3: Round 352 to the nearest hundred.	**400** (also accepted: four hundred)	352 is between 300 and 400. The tens digit is 5, so round up. 352 rounds to 400.
w2_s2	Set 3: Which text feature lists small topics in ABC order with page numbers, at the back of a book? [the index / the table of contents / a caption / a heading]	**the index**	The index is at the back, in ABC order. The table of contents is at the front, in page order.

MATH REVIEW: PICTURE GRAPHS AND BAR GRAPHS (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
m21a_s0	Set 1: Two steps! Use the jam picture graph. How many more plum and pear jars together were sold than apple jars?	**45** (also accepted: 45 jars, 45 more, 45 more jars)	Step 1: Plum is 6 × 10 = 60 and pear is 3 × 10 = 30, so 60 + 30 = 90 jars. Apple is 4 whole circles (40) and half a circle (5), so 45 jars. Step 2: 90 − 45 = 45 more jars.
m21b_s0	Set 1: Find the mistake. Wendell says the apple row shows 4 and a half jars, because it has 4 and a half circles. What went wrong? [He forgot the key. 4 circles × 10 = 40, and half a circle is 5, so the row shows 45 jars. / Nothing is wrong. The row shows 4 and a half jars. / The row shows 50 jars, because you always round up. / The row shows 14 jars, because 4 + 10 = 14.]	**He forgot the key. 4 circles × 10 = 40, and half a circle is 5, so the row shows 45 jars.**	Always read the key. Each circle is 10 jars, and half a circle is half of 10, which is 5. 40 + 5 = 45 jars.
m21a_s1	Set 2: Use the kite bar graph. How many fewer kites were sold on Wednesday than on Thursday?	**25** (also accepted: 25 kites, 25 fewer, 25 fewer kites)	Thursday stops halfway between 50 and 60, so it shows 55. Wednesday stops on 30. 55 − 30 = 25 fewer kites.
m21b_s1	Set 2: Which explanation is correct? Why does the Tuesday bar show 45 kites? [The scale counts by 10, and the bar stops halfway between 40 and 50. Halfway is 45. / The bar is about 4 and a half lines tall, so it shows 4 and a half kites. / The bar goes past 40, so you round up to 50. / The bar is the second bar, so it shows 4 + 5 = 45.]	**The scale counts by 10, and the bar stops halfway between 40 and 50. Halfway is 45.**	Read the scale first: 10, 20, 30, 40, 50. The bar stops right between 40 and 50, and the number halfway between them is 45.
m21a_s2	Set 3: Use the sun hat picture graph. How many sun hats were sold in all, counting all three rows?	**30** (also accepted: 30 hats, 30 sun hats)	Each circle is 2 hats, and half a circle is 1. Red: 5 × 2 = 10. Blue: 3 × 2 = 6, plus 1 = 7. Green: 6 × 2 = 12, plus 1 = 13. 10 + 7 + 13 = 30 hats.
m21b_s2	Set 3: Two steps! Use the same graph. How many more green hats than blue hats were sold? [6 more hats / 3 more hats / 20 more hats / 4 more hats]	**6 more hats**	Step 1: Green is 13 hats and blue is 7 hats. Step 2: 13 − 7 = 6 more green hats. Or: green has 3 more circles, and 3 × 2 = 6.

MATH REVIEW: MEASURING AND LINE PLOTS (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
m22a_s0	Set 1: How long is this paintbrush handle? Type your answer like 2 1/2 inches.	**3 1/4** (also accepted, with or without the unit word: 3 and 1/4, three 1/4, three and 1/4, 3 one fourth, 3 and one fourth, three one fourth, and 101 other spellings of the same answer)	The handle starts at 0, passes the 3, and stops at the first short mark after 3. That is 1/4 past 3, so it is 3 1/4 inches.
m22b_s0	Set 1: Find the mistake. Cyrus says this piece of chalk is 3 3/4 inches long. What went wrong? [The chalk starts at 1, not 0. From 1 to 3 3/4 is 2 3/4 inches. / Nothing is wrong. It ends at 3 3/4, so it is 3 3/4 inches. / It is 4 inches long, because you round up. / It is 3 inches long, because you only count whole inches.]	**The chalk starts at 1, not 0. From 1 to 3 3/4 is 2 3/4 inches.**	When an object does not start at 0, count from where it starts. From 1 to 3 is 2 inches, and then 3/4 more: 2 3/4 inches.
m22a_s1	Set 2: Use the ribbon line plot. How many pieces are longer than 2 inches?	**3** (also accepted: 3 pieces, three)	Longer than 2 means the lengths to the right of 2: 2 1/4 has 1 X and 2 1/2 has 2 X marks. 1 + 2 = 3 pieces.
m22b_s1	Set 2: Which explanation is correct? Which ribbon length is the most common? [2 inches, because it has the tallest stack of X marks: 5. / 2 1/2 inches, because it is the longest length. / 1 1/2 inches, because it is first on the line. / 2 1/4 inches, because it has only 1 X.]	**2 inches, because it has the tallest stack of X marks: 5.**	Most common means the length with the most X marks. 2 inches has 5 X marks, more than any other length.
m22a_s2	Set 3: Two steps! Barnaby's line plot shows 3 sticks at 4 1/4 inches, 5 sticks at 4 1/2 inches, and 2 sticks at 4 3/4 inches. He needs 12 sticks that are at least 4 1/2 inches long. How many more sticks must he cut?	**5** (also accepted: 5 sticks, 5 more, 5 more sticks, five)	Step 1: At least 4 1/2 means 4 1/2 or longer: 5 + 2 = 7 sticks. The 4 1/4 sticks are too short. Step 2: 12 − 7 = 5 more sticks.
m22b_s2	Set 3: Find the mistake. Zelda says 2/4 inch is longer than 1/2 inch, because 2 is more than 1. What went wrong? [2 of the 4 equal parts of an inch reach the middle mark, the same spot as 1/2 inch. They are the same length. / Nothing is wrong. 2/4 inch is longer. / 2/4 inch is shorter, because fourths are small. / 2/4 inch is 2 whole inches long.]	**2 of the 4 equal parts of an inch reach the middle mark, the same spot as 1/2 inch. They are the same length.**	Cut an inch into 4 equal parts. 2 of those parts fill half of the inch, so 2/4 inch and 1/2 inch end at the same mark.

MATH REVIEW: QUADRILATERALS AND EQUAL PARTS (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
m23a_s0	Set 1: A tile has 4 sides that are each 5 inches long. None of its corners are right angles. Type the best name for it: rectangle, rhombus, or square.	**rhombus** (also accepted: a rhombus)	All 4 sides are the same length, so it is a rhombus. It has no right angles, so it is not a rectangle or a square.

ID	Question	Answer	Why (explain to student)
m23b_s0	Set 1: Which explanation is correct? Ansel says a square is also a rectangle. Calla says a square cannot be a rectangle, because it does not have 2 long sides and 2 short sides. [Ansel is right. A rectangle only needs 4 right angles, and a square has 4 right angles. / Calla is right. A rectangle must have 2 long sides and 2 short sides. / Calla is right, because a square is a rhombus, so it cannot be anything else. / Neither is right. A square is not a quadrilateral.]	**Ansel is right. A rectangle only needs 4 right angles, and a square has 4 right angles.**	The rule for a rectangle is 4 right angles. A square has 4 right angles, so it is a special kind of rectangle.
m23a_s1	Set 2: Two steps! A rectangle tile is 7 inches long and 4 inches wide. How many inches of trim go around all 4 sides?	**22** (also accepted: 22 inches, 22 inch)	Step 1: In a rectangle, the sides across from each other match, so the sides are 7, 4, 7, and 4 inches. Step 2: 7 + 4 + 7 + 4 = 22 inches.
m23b_s1	Set 2: Find the mistake. Leopold cuts a clay tile into 3 parts. One part is big and two parts are small. He says each part is 1/3 of the tile. What went wrong? [The 3 parts are not the same size. Thirds must have equal area. / Nothing is wrong. Any 3 parts are thirds. / Each part is 1/2, because he made 2 cuts. / Each part is 1/4, because a tile has 4 sides.]	**The 3 parts are not the same size. Thirds must have equal area.**	A unit fraction like 1/3 only works when every part has the same area. Big and small parts are not thirds.
m23a_s2	Set 3: This tile is split into equal parts. What fraction of the tile's area is shaded? Type it like 1/2.	**1/5** (also accepted: 1 / 5, one fifth, one-fifth, a fifth)	The tile is split into 5 equal parts and 1 part is shaded, so the shaded part is 1/5 of the area.
m23b_s2	Set 3: Two steps! Henrietta has 2 square tiles and 3 triangle tiles. How many angles do all the tiles have in all? [17 angles / 20 angles / 15 angles / 5 angles]	**17 angles**	Step 1: 2 squares × 4 angles = 8, and 3 triangles × 3 angles = 9. Step 2: 8 + 9 = 17 angles.

MATH REVIEW: TWO-STEP PROBLEMS AND ROUNDING (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
m24a_s0	Set 1: Solve the equation: 8 × 6 − m = 29. What is m?	**19** (also accepted: m = 19, m=19, nineteen)	Step 1: Do the × part first: 8 × 6 = 48. Step 2: 48 − m = 29, so m = 48 − 29 = 19.
m24b_s0	Set 1: Find the mistake. Leopold solved 45 − 15 ÷ 5 = n. He got n = 6, because 45 − 15 = 30 and 30 ÷ 5 = 6. What went wrong? [He should divide first: 15 ÷ 5 = 3. Then 45 − 3 = 42, so n = 42. / Nothing is wrong. n = 6. / He should add: 45 + 15 + 5 = 65. / He should multiply: 15 × 5 = 75.]	**He should divide first: 15 ÷ 5 = 3. Then 45 − 3 = 42, so n = 42.**	When an equation has ÷ and −, do the ÷ part first. 15 ÷ 5 = 3, and 45 − 3 = 42.
m24a_s1	Set 2: Two steps! A bakery stand has 9 boxes with 7 buns in each box. They sell 38 buns. Let b stand for the buns left. What is b?	**25** (also accepted: b = 25, b=25, 25 buns, twenty-five)	Step 1: 9 × 7 = 63 buns. Step 2: 63 − 38 = 25, so b = 25 buns.

ID	Question	Answer	Why (explain to student)
m24b_s1	Set 2: Which explanation is correct? Cyrus adds 412 + 297 and gets 709. Is his answer reasonable? [Yes. 412 rounds to 400 and 297 rounds to 300. 400 + 300 = 700, and 709 is close to 700. / No. The estimate is 400 + 200 = 600. / No. You cannot check a sum by rounding. / Yes, because 709 is an odd number.]	**Yes. 412 rounds to 400 and 297 rounds to 300. 400 + 300 = 700, and 709 is close to 700.**	Round each number to the nearest hundred: 400 + 300 = 700. The exact answer, 709, is close to 700, so it is reasonable.
m24a_s2	Set 3: Solve the equation: k + 4 × 7 = 61. What is k?	**33** (also accepted: k = 33, k=33, thirty-three)	Step 1: Do the × part first: 4 × 7 = 28. Step 2: k + 28 = 61, so k = 61 − 28 = 33.
m24b_s2	Set 3: Which equation matches the story? Zelda has 6 bags with 8 rolls in each bag. Her family eats some rolls, and 30 rolls are left. Let e be the number of rolls they ate. [6 × 8 − e = 30 / 6 + 8 − e = 30 / 6 × 8 + 30 = e / e − 6 × 8 = 30]	**6 × 8 − e = 30**	She starts with 6 equal groups of 8, which is 6 × 8. Her family eats e rolls, so take away e. 30 rolls are left.

READING REVIEW: MAIN IDEA AND KEY DETAILS (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
r21a_s0	Set 1: *Read this text Honeybee Helpers Honeybees fly from flower to flower to drink a sweet juice called nectar. As they move, a yellow powder called pollen sticks to their fuzzy bodies. The pollen rubs off on the next flower, and that helps the plant make seeds. Many of the fruits we eat need this help. Honeybees are important helpers for plants.* In Honeybee Helpers, the main idea is the ___ sentence. Type first, middle, or last.	**last** (also accepted: the last, last sentence, the last sentence, last one)	Every other sentence explains how bees help plants. The last sentence, Honeybees are important helpers for plants, pulls them all together.
r21b_s0	Set 1: Find the mistake! Rosalind says the main idea is “Many of the fruits we eat need this help,” because it tells about food. What went wrong? [That is only one detail. All the sentences show how bees help plants, so the last sentence is the main idea. / Nothing went wrong. / The main idea is always the first sentence. / Honeybees do not visit flowers.]	**That is only one detail. All the sentences show how bees help plants, so the last sentence is the main idea.**	Ask which sentence ALL the others explain. The fruit sentence is one detail. Honeybees are important helpers for plants covers them all.
r21a_s1	Set 2: *Read this text Desert Tortoises A desert tortoise digs a burrow under the ground to hide from the hot sun. It can store water inside its body, so it can go a long time without a drink. Its thick shell protects it from animals that want to eat it.* “A desert tortoise has a shell” is too ___ to be the main idea of Desert Tortoises. Type big or small.	**small** (also accepted: too small)	It is only one detail. The text also tells about the burrow and storing water.
r21b_s1	Set 2: Desert Tortoises does not state its main idea. Which main idea is JUST RIGHT? [A desert tortoise has special ways to stay safe in the hot desert. / Many animals live in the world. / Tortoises dig holes. / The desert sun is hot.]	**A desert tortoise has special ways to stay safe in the hot desert.**	Just right covers every detail: the burrow, the stored water, and the shell. Many animals live in the world is too big. Tortoises dig holes is too small.

ID	Question	Answer	Why (explain to student)
r21a_s2	Set 3: *Read this text The Snowshoe Hare The snowshoe hare changes color to stay hidden. In summer, its fur is brown like the dirt and dead leaves. In fall, it starts to grow white fur. By winter, it is white like the snow. In spring, its brown fur grows back.* In The Snowshoe Hare, the main idea is the ___ sentence. Type first, middle, or last.	**first** (also accepted: the first, first sentence, the first sentence, first one)	The first sentence says the hare changes color to stay hidden. Every other sentence tells how its fur changes in each season.
r21b_s2	Set 3: Which explanation is correct? Cyrus says "Animals live where it snows" is just right for The Snowshoe Hare. Rosalind says it is too big. [Rosalind is right. It could fit a whole book and does not tell about the hare changing color. / Cyrus is right. Any true sentence about snow is just right. / Cyrus is right, because it is short. / Neither is right. It is too small.]	**Rosalind is right. It could fit a whole book and does not tell about the hare changing color.**	A just-right main idea tells what THIS text is mostly about: the snowshoe hare changes color to stay hidden.

READING REVIEW: TEXT FEATURES AND SEARCH TOOLS (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
r22a_s0	Set 1: *Read this page Field Guide: Rivers and Lakes Table of Contents: Rivers, 2 · Lakes, 6 · River Animals, 10 · Glossary, 14 · Index, 15 Index: beavers, 10, 12 · dams, 4, 11 · otters, 10 · waterfalls, 3* Use the index. Beavers are on page 10 and on page ___. Type the number.	**12** (also accepted: page 12, twelve)	The index says beavers, 10, 12. An index lists every page where a topic shows up.
r22b_s0	Set 1: Find the mistake! Wendell wants to know what the word estuary means, so he looks in the table of contents. What went wrong? [The table of contents lists sections and pages. The glossary on page 14 tells what words mean. / Nothing went wrong. / He should look at the index on page 15 for the meaning. / Estuary is a section title.]	**The table of contents lists sections and pages. The glossary on page 14 tells what words mean.**	Use the glossary for meanings. The table of contents and the index give page numbers, not meanings.
r22a_s1	Set 2: *Read this page Page 11: How Beavers Build Heading: How Beavers Build Beavers cut down small trees with their strong front teeth. They pile up sticks, rocks, and mud to build a dam across a stream. Caption: A beaver carries a branch to its dam. Sidebar, Fast Facts: A beaver can stay underwater for about 15 minutes.* Use the sidebar. A beaver can stay underwater for about ___ minutes. Type the number.	**15** (also accepted: 15 minutes, fifteen)	The Fast Facts sidebar says a beaver can stay underwater for about 15 minutes. A sidebar holds quick extra facts.

ID	Question	Answer	Why (explain to student)
r22b_s1	Set 2: Which explanation is correct? Imogen says the caption tells what the whole page is about. Barnaby says the heading, How Beavers Build, tells what the whole page is about. [Barnaby is right. A heading names the topic of the page. A caption only tells about the picture. / Imogen is right. A caption always tells about the whole page. / Imogen is right, because the caption is longer. / Neither is right. The sidebar tells what the page is about.]	**Barnaby is right. A heading names the topic of the page. A caption only tells about the picture.**	The heading is the title of a page or section. The caption, A beaver carries a branch to its dam, tells only about the picture.
r22a_s2	Set 3: *Read this page Field Guide: Rivers and Lakes Table of Contents: Rivers, 2 · Lakes, 6 · River Animals, 10 · Glossary, 14 · Index, 15 Index: bridges, 5 · fish, 7, 10, 13 · islands, 8 · waterfalls, 3* Two steps! Question: On how many pages can you read about fish? Find fish in the index, then count its pages. Type the number.	**3** (also accepted: 3 pages, three)	Step 1: The index says fish, 7, 10, 13. Step 2: Count the page numbers: 7, 10, and 13 make 3 pages.
r22b_s2	Set 3: Which question would the GLOSSARY answer? [What does the word marsh mean? / Which pages tell about fish? / On which page does the Lakes section start? / What does the picture on page 8 show?]	**What does the word marsh mean?**	The glossary gives meanings. The index and the table of contents give pages, and a caption tells what a picture shows.

READING REVIEW: POINT OF VIEW (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
r23a_s0	Set 1: *Read this story Skating Day I zipped up my coat and ran down to the frozen pond. Grandpa Ansel was already there with two pairs of skates. "Ready?" he asked. I could not stop smiling. I think skating is the best part of winter!* Skating Day is told in ___ person. Type first or third.	**first** (also accepted: first person, 1st)	The narrator says I and my outside the quotation marks, so the narrator is a character. That is first person.
r23b_s0	Set 1: Find the mistake! Zelda says Skating Day is told in third person, because it has the name Grandpa Ansel in it. What went wrong? [Names can be in any story. The narrator tells about herself with I and my, so it is first person. / Nothing went wrong. / Every story with a name is third person. / Skating Day has no narrator.]	**Names can be in any story. The narrator tells about herself with I and my, so it is first person.**	Look at how the narrator talks. I zipped up my coat uses I and my, so Skating Day is first person, even with a name in it.
r23a_s1	Set 2: *Read this story The Tall Slide Calla stared up at the tall slide. Her knees shook. "I am not sure about this," she whispered. Her brother Leopold zoomed down and laughed. "This is the best slide in the whole park!" he shouted.* Which character thinks the slide is the best? Type the name.	**Leopold**	Leopold shouts, "This is the best slide in the whole park!" That is his point of view. Calla feels nervous.

ID	Question	Answer	Why (explain to student)
r23b_s1	Set 2: Which explanation is correct? Ansel says The Tall Slide is told in first person, because Calla says I. Imogen says it is told in third person, because the narrator uses she, her, and names, and the word I is inside quotation marks. [Imogen is right. I inside quotation marks is a character talking. The narrator is outside the story. / Ansel is right. Any story with the word I is first person. / Ansel is right, because Calla is the narrator. / Neither is right. The story has no narrator.]	**Imogen is right. I inside quotation marks is a character talking. The narrator is outside the story.**	Skip the quotation marks. The narrator says Calla stared and her knees shook, so the narrator is outside the story: third person.
r23a_s2	Set 3: *Read this story Rainy Saturday Barnaby thought the rainy day was boring. He sighed and pressed his nose to the window. His sister Imogen grabbed her boots. "Puddles!" she cheered, and she ran outside to splash.* How does Imogen feel about the rain? Type one word: happy, bored, or scared.	**happy** (also accepted: excited, glad)	Imogen cheers "Puddles!" and runs outside to splash. Her words and actions show she is happy about the rain. Barnaby is the one who feels bored.
r23b_s2	Set 3: Which sentence shares a READER's own point of view AND gives a reason? [I would splash with Imogen, because jumping in puddles is fun. / Barnaby thinks the rain is boring. / Imogen grabbed her boots. / I like rain.]	**I would splash with Imogen, because jumping in puddles is fun.**	A reader's point of view tells what the READER thinks and gives a reason after because. I like rain has no reason, and the others tell about the characters.

READING REVIEW: BIOGRAPHY, SEQUENCE, AND CAUSE AND EFFECT (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
r24a_s0	Set 1: *Read this biography Mae Jemison: Reaching for the Stars Mae Jemison was born on October 17, 1956, in Decatur, Alabama. When she was a young child, her family moved to Chicago, Illinois, so Mae grew up there. She loved science, and she was curious about the stars. In 1973, when she was 16 years old, Mae started college at Stanford University in California. Later, she became a doctor. In 1987, NASA chose her to be an astronaut, so she began years of training. On September 12, 1992, she flew into space on the space shuttle Endeavour. She was the first African American woman to travel in space.* Mae Jemison flew into space in the year ___.	**1992**	The biography says: On September 12, 1992, she flew into space on the space shuttle Endeavour.
r24b_s0	Set 1: Which list is in the correct order? [Born in Alabama → starts college → NASA chooses her → flies into space / Starts college → born in Alabama → flies into space → NASA chooses her / NASA chooses her → born in Alabama → starts college → flies into space / Flies into space → NASA chooses her → starts college → born in Alabama]	**Born in Alabama → starts college → NASA chooses her → flies into space**	Put the smallest year first: 1956 (born), 1973 (college), 1987 (NASA chooses her), 1992 (flies into space).

ID	Question	Answer	Why (explain to student)
r24a_s1	Set 2: *Same passage as Set 1.* Two steps! How old was Mae when she flew into space on September 12, 1992? Hint: her birthday is in October. Type the number.	**35** (also accepted: 35 years old, 35 years, thirty-five)	Step 1: 1992 − 1956 = 36. Step 2: September comes before her October birthday, so she had not turned 36 yet. She was 35.
r24b_s1	Set 2: Find the mistake! Rosalind says this text is an autobiography, because it tells a true story about a real life. What went wrong? [Someone else wrote it, using she, her, and Mae's name, so it is a biography. / Nothing went wrong. / It is fiction, because it tells a story. / An autobiography is always about astronauts.]	**Someone else wrote it, using she, her, and Mae's name, so it is a biography.**	A biography and an autobiography are both true. A biography is written by someone else in third person, like this one.
r24a_s2	Set 3: *Same passage as Set 1.* NASA chose Mae to be an astronaut. What was the EFFECT? She began years of ___. Type one word.	**training**	The sentence says NASA chose her to be an astronaut, so she began years of training. So comes right before the effect.
r24b_s2	Set 3: Which explanation is correct? Cyrus says the cause in the first paragraph is that Mae grew up in Chicago. Zelda says the cause is that her family moved to Chicago, and the effect is that Mae grew up there. [Zelda is right. The word so comes right before the effect, Mae grew up there. / Cyrus is right. Growing up in Chicago caused the move. / Cyrus is right, because Chicago is a city. / Neither is right. The paragraph has no cause and effect.]	**Zelda is right. The word so comes right before the effect, Mae grew up there.**	The family moved first. That is WHY Mae grew up in Chicago. The move is the cause, and growing up there is the effect.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match key	**tells what each picture stands for**	2 points if matched on the first try.
match_s0_2	Set 1: Match scale	**the numbers a bar graph counts by**	2 points if matched on the first try.
match_s0_3	Set 1: Match line plot	**a number line with an X for each measurement**	2 points if matched on the first try.
match_s0_4	Set 1: Match rhombus	**a quadrilateral with 4 equal sides**	2 points if matched on the first try.
match_s0_5	Set 1: Match estimate	**a close answer found with easy numbers**	2 points if matched on the first try.
match_s1_1	Set 2: Match main idea	**the most important point**	2 points if matched on the first try.
match_s1_2	Set 2: Match index	**topics in ABC order at the back of a book**	2 points if matched on the first try.
match_s1_3	Set 2: Match narrator	**the voice that tells the story**	2 points if matched on the first try.
match_s1_4	Set 2: Match biography	**a true life story written by someone else**	2 points if matched on the first try.

ID	Question	Answer	Why (explain to student)
match_s1_5	Set 2: Match effect	**what happens because of a cause**	2 points if matched on the first try.
match_s2_1	Set 3: Match glossary	**word meanings at the back of a book**	2 points if matched on the first try.
match_s2_2	Set 3: Match first person	**the narrator says I, me, and my**	2 points if matched on the first try.
match_s2_3	Set 3: Match cause	**why something happens**	2 points if matched on the first try.
match_s2_4	Set 3: Match rectangle	**a quadrilateral with 4 right angles**	2 points if matched on the first try.
match_s2_5	Set 3: Match quarter inch	**one of 4 equal parts of an inch**	2 points if matched on the first try.

FIX THE SENTENCE (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
fs1_s0	Set 1: Fix the sentence: rosalind and cyrus visited seattle washington	**Rosalind and Cyrus visited Seattle, Washington.**	Rosalind and Cyrus are names, and Seattle and Washington are names of places, so they all get capital letters. Put a comma between the city and the state, and end with a period.
fs2_s0	Set 1: Fix the sentence: cyrus drew squares rhombuses and rectangles	**Cyrus drew squares, rhombuses, and rectangles.** (also accepted: Cyrus drew squares, rhombuses and rectangles.)	Cyrus is a name and starts the sentence, so it gets a capital C. Put commas between the things in the list and end with a period.
fs1_s1	Set 2: Fix the sentence: did roberto clemente play for pittsburgh	**Did Roberto Clemente play for Pittsburgh?**	Start with a capital D. Roberto Clemente is a name and Pittsburgh is a city, so they get capital letters. This sentence asks something, so it ends with a question mark.
fs2_s1	Set 2: Fix the sentence (one verb is wrong, and there are two graphs): the two bar graphs was finished	**The two bar graphs were finished.**	Start with a capital T. Two bar graphs means more than one, so use were, not was. End with a period.
fs1_s2	Set 3: Fix the sentence: rosalind read about mammoth cave in kentucky	**Rosalind read about Mammoth Cave in Kentucky.**	Rosalind is a name, Mammoth Cave is the name of a place, and Kentucky is a state, so they get capital letters. A telling sentence ends with a period.
fs2_s2	Set 3: Fix the sentence (one verb is wrong, and there are two people): imogen and wendell sorts the tiles	**Imogen and Wendell sort the tiles.**	Imogen and Wendell are names, so they get capital letters. Two people means use sort, not sorts. End with a period.

TYPE-IT CORRAL (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
t1_s0	Set 1: Two steps! On a picture graph, each picture stands for 5 votes. The pizza row has 7 pictures and the taco row has 4 pictures. How many votes are there in all?	**55** (also accepted: 55 votes, fifty-five)	Step 1: Pizza is 7 × 5 = 35 votes and taco is 4 × 5 = 20 votes. Step 2: 35 + 20 = 55 votes. Or: 7 + 4 = 11 pictures, and 11 × 5 = 55.
t2_s0	Set 1: The voice that tells a story is called the ___.	**narrator** (also accepted: the narrator)	The narrator is the voice that tells the story. It can be a character (first person) or a voice outside the story (third person).
t1_s1	Set 2: Two steps! Cyrus glues 3 pieces of wood that are each 1/4 inch long onto the end of a 2 1/4 -inch block. How long is the new piece? Type your answer like 5 inches.	**3** (also accepted: 3 inches, 3 inch, 3 in, three, three inches)	Step 1: 1/4 + 1/4 + 1/4 = 3/4 inch. Step 2: 2 1/4 + 3/4 = 3 inches, because 1/4 and 3/4 make 1 whole inch.
t2_s1	Set 2: The true story of a real person's life, written by someone else, is called a ___.	**biography** (also accepted: a biography)	A biography is written by someone else, in third person. A life story a person writes about themself is an autobiography.
t1_s2	Set 3: Round 763 to the nearest hundred.	**800** (also accepted: eight hundred)	763 is between 700 and 800. The tens digit is 6, which is 5 or more, so round up. 763 rounds to 800.
t2_s2	Set 3: The list of word meanings in ABC order at the back of a book is the ___.	**glossary** (also accepted: the glossary)	The glossary gives word meanings. The index, also at the back, gives page numbers.

BOSS BATTLE: THE DATA AND DESIGN FAIR (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
bb1_s0	Set 1: *Rosalind and Cyrus's flyer The Data and Design Fair Come to the Data and Design Fair this Saturday in the school gym! Our class made 6 tile posters with 8 tiles on each poster. Each tile is a square cut into 4 equal parts. There will be tables of graphs, ruler games, and a reading corner. I think the tile posters are the most beautiful part of the fair!* Two steps! Before the fair, 9 tiles fell off the posters. Let t stand for the tiles still on the posters. What is t?	**39** (also accepted: t = 39, t=39, 39 tiles, thirty-nine)	Step 1: 6 posters × 8 tiles = 48 tiles. Step 2: 48 − 9 = 39, so t = 39 tiles.
bb2_s0	Set 1: Which explanation is correct? The flyer says, "I think the tile posters are the most beautiful part of the fair!" Rosalind says this sentence shares the writer's point of view. Cyrus says it is a fact that can be proved. [Rosalind is right. The words I think show what the writer thinks and feels. / Cyrus is right. Every sentence on a flyer is a fact. / Cyrus is right, because the posters are real. / Neither is right. The sentence is a caption.]	**Rosalind is right. The words I think show what the writer thinks and feels.**	A point of view is how someone thinks or feels. I think and most beautiful show the writer's feelings, and other people might feel differently.

ID	Question	Answer	Why (explain to student)
bb1_s1	Set 2: *Rosalind and Cyrus's flyer The Tile Art Table Visit the Tile Art Table at the Data and Design Fair! Our rhombus tiles have 4 sides, and each side is 3 inches long. Our square tiles are each cut into 4 equal parts. We measured every tile frame to the nearest quarter inch and made a line plot of the lengths.* Two steps! Calla glues ribbon around all 4 sides of 2 rhombus tiles. How many inches of ribbon does she need in all?	**24** (also accepted: 24 inches, 24 inch, twenty-four)	Step 1: A rhombus has 4 equal sides, so one tile needs 3 + 3 + 3 + 3 = 12 inches. Step 2: 2 tiles need 12 + 12 = 24 inches.
bb2_s1	Set 2: Which explanation is correct? Zelda cuts a square tile into 4 equal parts with 2 straight cuts. Ansel says each part is 1/4 of the tile. Zelda says each part is 1/2, because she made 2 cuts. [Ansel is right. Count the equal parts, not the cuts. 4 equal parts means each is 1/4. / Zelda is right. 2 cuts means halves. / Zelda is right, because a square has 2 long sides. / Neither is right. Each part is 1/3.]	**Ansel is right. Count the equal parts, not the cuts. 4 equal parts means each is 1/4.**	The bottom number tells how many equal parts make the whole. 2 cuts made 4 equal parts, so each part is 1/4.
bb1_s2	Set 3: *Rosalind and Cyrus's flyer The Reading Corner Stop by the Reading Corner at the Data and Design Fair! Our class wrote a biography of Mae Jemison and a booklet called Prairie Partners. The booklet has a table of contents, a glossary, and an index. The corner has 4 shelves with 9 books on each shelf.* Two steps! Kids borrow 17 books from the shelves. Let b stand for the books left. What is b?	**19** (also accepted: b = 19, b=19, 19 books, nineteen)	Step 1: 4 shelves × 9 books = 36 books. Step 2: 36 − 17 = 19, so b = 19 books.
bb2_s2	Set 3: Find the mistake. Cyrus says the class biography of Mae Jemison is told in first person, because it is about one person. What went wrong? [A biography is written by someone else, so it uses she, her, and Mae's name. It is third person. / Nothing went wrong. / A biography is always first person. / A biography is fiction, so it has no narrator.]	**A biography is written by someone else, so it uses she, her, and Mae's name. It is third person.**	Being about one person does not make a text first person. A biography tells someone's life from outside, in third person.

BOSS BATTLE: THE LAST PANEL (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
bb3_s0	Set 1: Rosalind counted visitors and made this bar graph. How many more visitors came at 10 a.m. than at 9 a.m.?	**25** (also accepted: 25 visitors, 25 more, 25 more visitors, twenty-five)	The 10 a.m. bar stops on 60. The 9 a.m. bar stops halfway between 30 and 40, so it shows 35. 60 − 35 = 25 more visitors.

ID	Question	Answer	Why (explain to student)
bb4_s0	Set 1: Find the mistake. Cyrus says 50 visitors came at 11 a.m., because the bar goes past the 40 line. What went wrong? [The bar stops halfway between 40 and 50, so it shows 45 visitors. / Nothing is wrong. 50 visitors came. / The bar shows 4 and a half visitors. / The bar shows 40 visitors, because you always round down.]	**The bar stops halfway between 40 and 50, so it shows 45 visitors.**	Read where the bar stops, not just which line it passes. Halfway between 40 and 50 is 45.
bb3_s1	Set 2: Use the tile frame line plot. How many frames are 5 1/2 inches or longer?	**8** (also accepted: 8 frames, eight)	5 1/2 or longer means 5 1/2, 5 3/4, and 6: 4 + 3 + 1 = 8 frames.
bb4_s1	Set 2: Find the mistake. Leopold says only 4 frames were measured, because there are 4 lengths on the line plot. What went wrong? [Each X is one frame. 2 + 4 + 3 + 1 = 10 frames were measured. / Nothing is wrong. 4 frames were measured. / 6 frames were measured, because 6 is the longest length. / 3 frames were measured, because 5 3/4 has 3 X marks.]	**Each X is one frame. 2 + 4 + 3 + 1 = 10 frames were measured.**	On a line plot, every X stands for one measured thing. Count all the X marks: 10 frames.
bb3_s2	Set 3: Use the picture graph. How many books were borrowed in all on the three days?	**31** (also accepted: 31 books, thirty-one)	Each circle is 2 books, and half a circle is 1. Monday: 4 × 2 = 8. Tuesday: 6 × 2 = 12, plus 1 = 13. Wednesday: 5 × 2 = 10. 8 + 13 + 10 = 31 books.
bb4_s2	Set 3: Which explanation is correct? Rosalind adds 286 + 197 and gets 483. Is her answer reasonable? [Yes. 286 rounds to 300 and 197 rounds to 200. 300 + 200 = 500, and 483 is close to 500. / No. The estimate is 200 + 100 = 300. / No. You cannot round to check a sum. / Yes, because 483 has three digits.]	**Yes. 286 rounds to 300 and 197 rounds to 200. 300 + 200 = 500, and 483 is close to 500.**	Round each number to the nearest hundred: 300 + 200 = 500. The exact sum, 483, is close to 500, so it is reasonable.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade3/prepost/KCE_Checkpoint_PrePost_G3W25.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Each flower stands for 10 bouquets; one row has 3 and a half flowers	35 bouquets. 3 × 10 = 30, and half a flower is half of 10, which is 5. 30 + 5 = 35.
2	Draw a line 2 1/4 inches long, starting at 0	The line starts at 0 and ends at the first quarter-inch mark after 2. Check with a ruler.
3	A tile has 4 right angles and 4 sides that are each 6 inches long	square. It has 4 right angles (a rectangle) and 4 equal sides (a rhombus), so square is the most exact name.
4	Round 538 to the nearest hundred	500. The tens digit is 3, which is less than 5, so round down.
5	“My dog Pepper wagged her tail when I got home.” First or third person?	First person. Circle My or I (either is correct). Her tells about the dog, not the narrator.
6	Which tool tells you what a word means?	the glossary

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	7 boxes of 6 muffins; 15 left; s = muffins sold	Equation: 7 × 6 − s = 15. Step 1: 7 × 6 = 42. Step 2: 42 − s = 15, so s = 42 − 15 = 27 muffins sold. Accept any correct equation, such as 42 − s = 15.
2	Wendell says a bar halfway between 20 and 30 shows 30	It shows 25. The scale counts by 10, and the number halfway between 20 and 30 is 25. The bar does not reach the 30 line.
3	Line plot: 2 pegs at 3 1/4, 5 at 3 1/2, 1 at 3 3/4. How many longer than 3 1/4 inches?	6 pegs. Longer than 3 1/4 means 3 1/2 and 3 3/4: 5 + 1 = 6. The 2 pegs at 3 1/4 do not count.
4	Fox text (tail over its nose; sleeps in a sunny spot on cold days): write the main idea	Model: The fox has ways to stay warm on cold days. The main idea is not stated; accept any just-right sentence that covers both details (not just the tail or just the sunny spot).

★ CHALLENGE

#	Prompt	Model answer / grading note
1	Rectangle split into 4 equal parts; shade 1 part	1/4 (one fourth). Drawing shows 4 parts with equal area and 1 shaded; the parts may be strips or other equal-area shapes.